

LAB EXPERIMENT RECORD

SET 8: EMPLOYEE PERFORMANCE ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	B KIRTHANAA
Register Number	192324244
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Aim

To analyze employee performance data using a line chart, a bar chart and a performance table in R, and to build an interactive dashboard using Tableau. Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

PROGRAM 1: Line Chart of Performance Trend

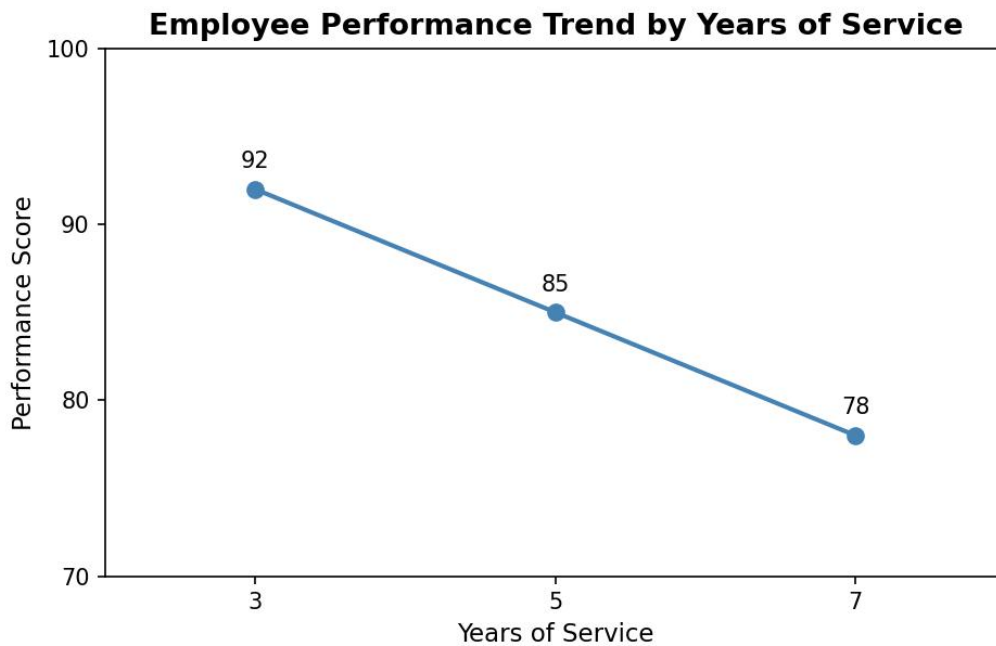
Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
# Dataset
employee <- data.frame(
  Employee_ID      = 1:3,
  Department       = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Arrange by years of service so the trend runs over time
emp_sorted <- employee[order(employee$Years_of_Service), ]

# Line Chart
plot(emp_sorted$Years_of_Service, emp_sorted$Performance_Score,
     type = "b", pch = 19, lwd = 2, col = "steelblue",
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Employee Performance Trend by Years of Service",
     xlim = c(2, 8), ylim = c(70, 100), xaxt = "n")
axis(1, at = emp_sorted$Years_of_Service)
text(emp_sorted$Years_of_Service, emp_sorted$Performance_Score,
     labels = emp_sorted$Performance_Score, pos = 3)
```

OUTPUT:



Output 1: Employee Performance Trend (Line Chart)

Observation: Performance falls steadily as years of service increase. The score drops from 92 at 3 years of service to 85 at 5 years and 78 at 7 years, a fall of 7 points at each step, or about 3.5 points per year. The employee with the shortest service (HR) has the highest score and the one with the longest service (Marketing) has the lowest.

PROGRAM 2: Bar Chart of Employees by Department

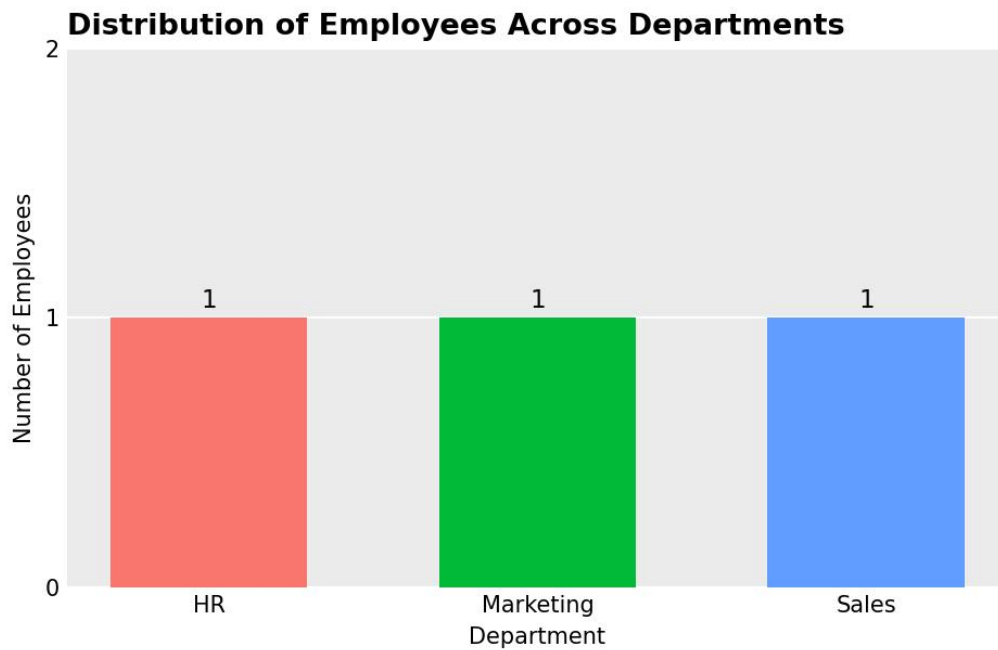
Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
library(ggplot2)

# Count of employees in each department
dept_count <- as.data.frame(table(employee$Department))
colnames(dept_count) <- c("Department", "Employees")

# Bar Chart
ggplot(dept_count, aes(x = Department, y = Employees, fill = Department)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = Employees), vjust = -0.5) +
  scale_y_continuous(breaks = 0:2, limits = c(0, 2)) +
  labs(title = "Distribution of Employees Across Departments",
       x = "Department",
       y = "Number of Employees",
       fill = "Department")
```

OUTPUT:



Output 2: Distribution of Employees Across Departments (Bar Chart)

Observation: The dataset contains 3 employees spread across 3 departments: HR, Marketing and Sales. Each department has exactly one employee, so all three bars are the same height and the distribution is perfectly even.

PROGRAM 3: Table of Employee Performance Data

Build a table to display the performance data for each employee. Label the table elements.

```
# Dataset
employee <- data.frame(
  Employee_ID      = 1:3,
  Department       = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Add a rating column based on the performance score
employee$Rating <- ifelse(employee$Performance_Score >= 90, "Excellent",
  ifelse(employee$Performance_Score >= 80, "Good", "Average"))

# Display the table
print(employee)

# Average performance score of all employees
mean(employee$Performance_Score)
```

Console Output

```

Employee_ID Department Years_of_Service Performance_Score Rating
1           1      Sales              5              85      Good
2           2        HR              3              92    Excellent
3           3   Marketing              7              78      Average
[1] 85

```

Output 3a: Employee Performance Table (R Console)

Employee ID	Department	Years of Service	Performance Score	Rating
1	Sales	5	85	Good
2	HR	3	92	Excellent
3	Marketing	7	78	Average
Average			85	

Output 3b: Employee Performance Table with Ratings

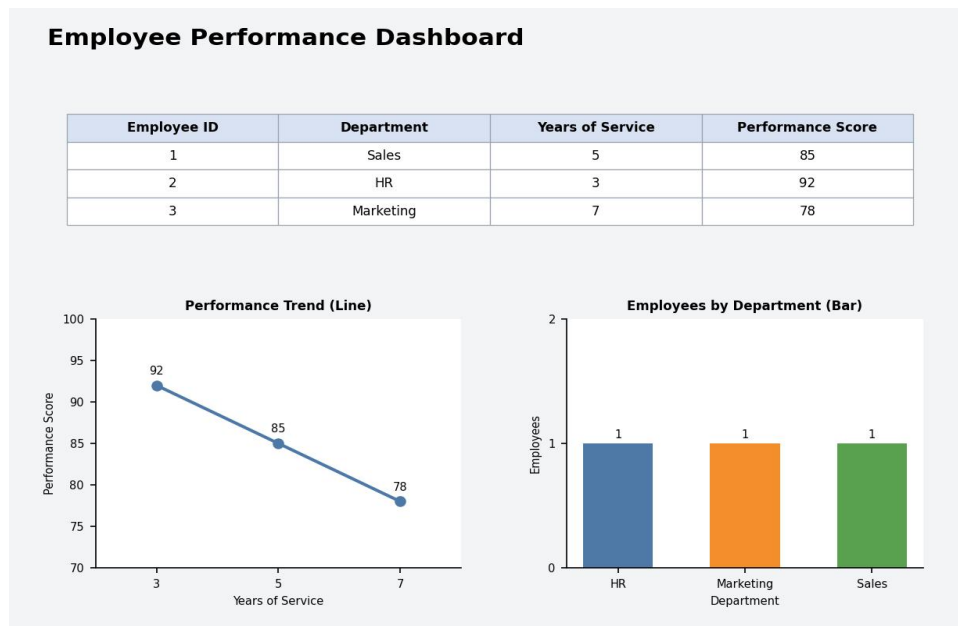
Observation: The table gives the exact performance score of each employee along with department and years of service. Employee 2 (HR, 3 years) has the highest score of 92 and is rated Excellent, Employee 1 (Sales, 5 years) scores 85 and is rated Good, and Employee 3 (Marketing, 7 years) has the lowest score of 78 and is rated Average. The average score is 85 and the gap between the highest and lowest score is 14 points.

PROGRAM 4: Interactive Employee Performance Dashboard in Tableau

Develop a Tableau dashboard combining the line chart, bar chart and the table for interactive exploration of employee performance data.

Steps in Tableau

1. Save the data as an Excel or CSV file with the columns Employee ID, Department, Years of Service and Performance Score, and connect to it in Tableau (Connect > Microsoft Excel / Text file).
2. Sheet 1 (Line Chart): Drag Years of Service to Columns and Performance Score to Rows (SUM). Change the Marks type to Line, add Performance Score to Label, and rename the sheet "Performance Trend".
3. Sheet 2 (Bar Chart): Drag Department to Columns and Employee ID to Rows, then change the measure to Count (Distinct). Drag Department to Color, add the count to Label, and rename the sheet "Employees by Department".
4. Sheet 3 (Table): Drag Employee ID, Department and Years of Service to Rows and Performance Score to Text. Rename the sheet "Performance Table".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag the three sheets onto the dashboard area.
6. Add interactivity: select each view and click Use as Filter, so clicking a department or an employee in one view filters the others. Add a Department filter (Show Filter > Apply to Worksheets > All Using This Data Source).
7. Add the title "Employee Performance Dashboard" and test the filters in Presentation Mode.



Output 4: Interactive Employee Performance Dashboard (Tableau)

Observation: The dashboard shows the performance trend, the department distribution and the employee performance table together. Clicking a department or an employee in one view filters the other views.

Procedure

1. Define the employee data (ID, department, years of service and performance score) in R as a data frame.
2. Sort the data by years of service and use `plot()` with `type = "b"` to draw the line chart of the performance trend.
3. Count the employees in each department with `table()` and use `ggplot2` with `geom_col()` to draw the bar chart.
4. Add a Rating column, display the data frame with `print()` and find the average with `mean()`.
5. Load the employee file into Tableau and build the line chart, bar chart and table worksheets.
6. Combine the worksheets in a Tableau dashboard and add filters and actions for interactivity.

RESULT:

The employee performance data was successfully analyzed using a line chart, a bar chart and a performance table in R. An interactive Tableau dashboard was also created, combining all three views so that users can explore employee performance by department and by years of service.

Key Functions and Tools Used

Function / Tool	Purpose
<code>data.frame()</code>	Creates the employee dataset and table.
<code>plot(type = "b")</code>	Creates the line chart with points.
<code>table()</code> / <code>ggplot()</code> / <code>geom_col()</code>	Counts employees per department and draws the bar chart.
<code>print()</code> / <code>mean()</code>	Displays the performance table and calculates the average score.
Tableau Dashboard	Combines worksheets with filters and actions.